

Agentic Voice AI Scheduler for Medical Appointment Booking

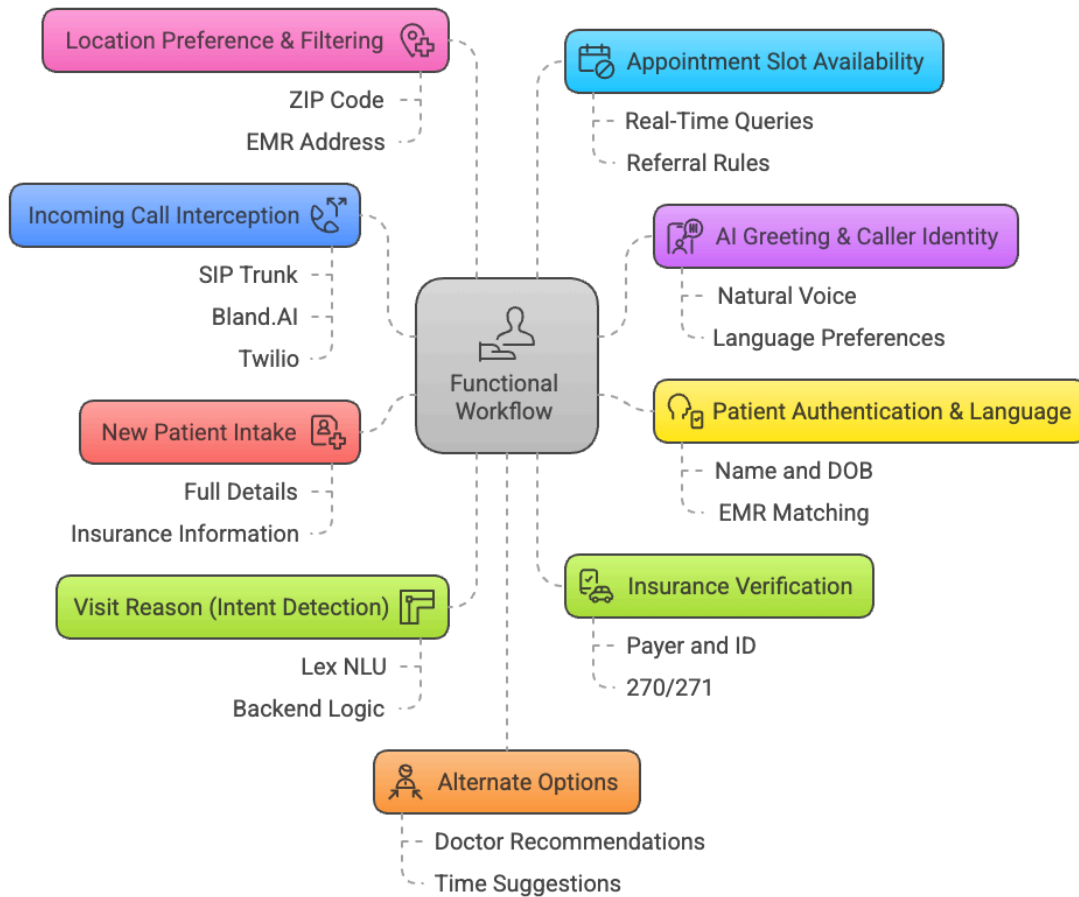
Project Overview

Brightcone.ai proposes the development of a fully agentic, AI-powered medical appointment scheduling system. This system will offer natural, human-like voice interaction for patients calling clinics or hospitals, supporting scheduling, rescheduling, cancellation, and escalation needs. Built using Amazon Lex and Bland.AI for voice AI, and leveraging AWS-native services for security and scalability, the system will integrate tightly with EMRs (e.g., Epic, eClinicalWorks) and insurance providers to streamline workflows and reduce human overhead.

Goals & Capabilities

- Automate end-to-end patient scheduling via voice and telephony
- Support returning and new patients with EMR and profile creation flows
- Detect patient intent in real-time and route to appropriate departments
- Verify insurance in real-time through integrated APIs (FHIR or 270/271)
- Handle location and time preferences dynamically
- Escalate to live human agents when needed with full context handoff
- Confirm appointments, send follow-up notifications, and gather pre-visit data
- Maintain compliance with HIPAA and data protection for best practices

AI-Driven Patient Call Management Workflow



Functional Workflow

1. Incoming Call Interception

- ❖ The patient calls the clinic.
- ❖ AI intercepts via SIP trunk through Bland.AI or Twilio.

2. AI Greeting & Caller Identity

- ❖ Natural voice welcome.
- ❖ Asks if the appointment is for self or others.
- ❖ Determines language preferences.

3. Patient Authentication & Language

- ❖ Name and DOB are collected.
- ❖ Matches EMR.
- ❖ Confirms identity via demographic.
- ❖ Auto-switch to preferred language.

4. **New Patient Intake**

- ❖ If unmatched then collects full details including insurance.
- ❖ Option to escalate or send form via SMS/email.

5. **Visit Reason (Intent Detection)**

- ❖ AI detects the type of visit using Lex NLU.
- ❖ Maps to the proper department using backend logic.

6. **Insurance Verification**

- ❖ Payer and ID collected.
- ❖ Performs real-time eligibility check via 270/271.
- ❖ Returns in-network status, copay, referral requirement.

7. **Location Preference & Filtering**

- ❖ Suggests nearby clinics based on ZIP or EMR address.
- ❖ Applies time/day preferences.

8. **Appointment Slot Availability**

- ❖ Queries EMR in real-time.
- ❖ Applies filters and shows available slots. Handles rules (referrals, specialties).

9. **Alternate Options**

- ❖ If slots declined then recommend other doctors or times.
- ❖ Offers to remember preferences.

10. **Escalation (if needed)**

- ❖ Transfers to a human agent via queue or callback.
- ❖ Logs context and transcript.

11. **Confirmation & Booking**

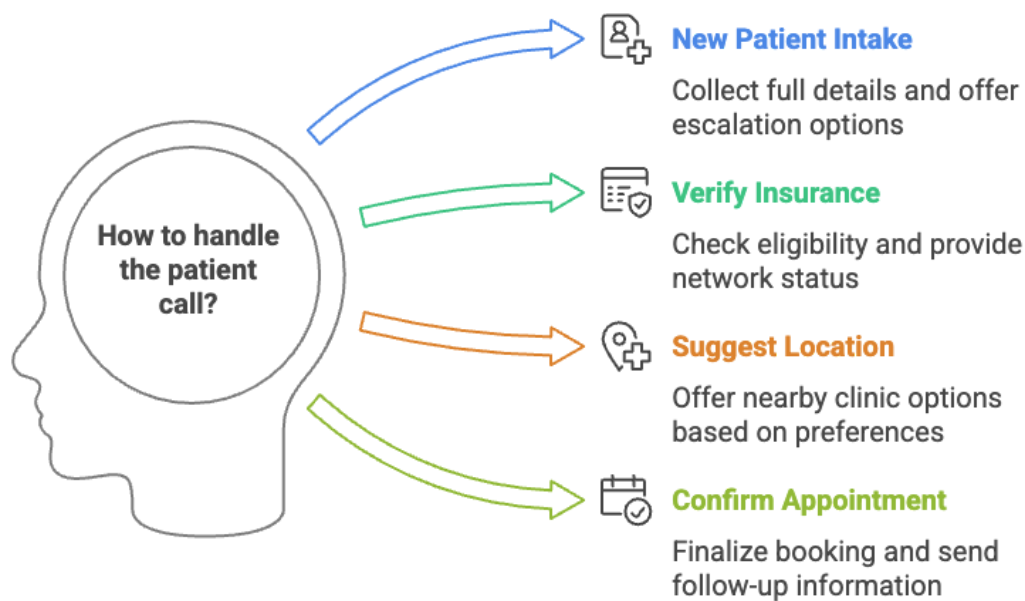
- ❖ Confirms verbally and logs appointments in EMR.
- ❖ Captures all visit metadata and context.

12. Post-Call Follow-up

- ❖ Send SMS/email, calendar link, directions, pre-visit forms.

13. End Call

- ❖ Confirms if any other help is needed. Signs off with a natural tone.



Deliverables (By Phase)

Phase 1: Core Development & Beta Delivery (Weeks 1–12)

Goal: Build and deliver a functional beta version of the Agentic Voice AI Scheduler, integrated with core EMR and scheduling workflows.

Weeks 1–2: Project Kickoff & Discovery

- Finalize project scope and success criteria.
- Confirm technology stack and system architecture
- Set up project tools and kickoff meetings

Weeks 3–4: Infrastructure Setup

- Configure AWS infrastructure (Lex, Lambda, API Gateway, S3)
- Deploy initial backend project (FastAPI)

Weeks 5–6: Pre-Development Setup

- Build backend skeleton with PostgreSQL schema
- Implement authentication and user management
- Set up CI/CD pipeline and API routing

Weeks 7–8: Core Features Development I

- Implement caller flow: greeting, language selection, identity handling
- Develop new patient intake and visit reason NLU logic
- Build initial integration for EPIC API

Weeks 9–10: Core Features Development II

- Implement real-time insurance eligibility checks via 270/271
- Build appointment lookup and scheduling with EMR integration

Weeks 11–12: EMR and Insurance Integration

- Complete EMR integrations (EPIC, eClinicalWorks)
- Finalize API certifications
- Deliver beta version to selected clinics for internal testing

Phase1 Outcome: A functioning Beta Version with working appointment booking, EMR integration, and patient flow logic.

Phase 2: Expansion, Compliance & Final Launch (Weeks 13–24)

Goal: Expand system capabilities, implement compliance and multilingual features, finalize QA, and support full rollout.

Weeks 13–14: Smart Routing and Fallback Logic

- Implement smart routing from intent to departments
- Add multilingual fallback and escalation pathways (Spanish, Chinese)

Weeks 15–16: QA and UAT Testing

- Conduct QA (manual + automated) on voice flows
- Run User Acceptance Testing (UAT) with beta clinics
- Resolve critical bugs

Weeks 17–18: SaaS Admin Portal & Analytics

- Build SaaS Admin Portal for clinics
- Integrate call analytics dashboard

Weeks 19–20: Compliance & Multilingual Expansion

- Enforce HIPAA/HITRUST compliance across infrastructure
- Finalize multilingual voice support for all key flows

Weeks 21–22: Final QA, Rollout & Beta(3 months)

- Run pre-launch testing and validation
- Roll out to pilot clinics for 3-month trial

Weeks 23–24: Hypercare & Optimization

- Provide hypercare support: monitor, debug, and optimize
- Gather post-launch feedback and performance metrics

Phase2 Outcome: Full system Launch, multilingual capability, compliance-ready backend, and post-rollout support.

Technology Stack

The Agentic Voice AI Scheduler will be developed using a robust and secure cloud-native stack, tailored specifically for healthcare workflows. The technologies are chosen to ensure scalability, real-time responsiveness, HIPAA compliance, and seamless integration with EMRs and telephony systems.

Voice Interface: We will use Amazon Lex to enable real-time intent detection through conversational AI, allowing the system to understand patient goals such as booking, rescheduling, or canceling appointments.

- Amazon Polly will synthesize natural-sounding voice responses in multiple languages (English, Spanish, Chinese, Telugu), enhancing accessibility.

- Amazon Transcribe will convert incoming patient speech into structured text, powering the real-time dialogue system.

Telephony Integration

- The system will connect with the clinic's phone lines using Bland.AI, a SIP-based telephony interface that enables full-duplex communication.
- For optional fallback and SMS workflows (e.g., appointment reminders, escalations), Twilio will be integrated where appropriate.

Backend System

- The backend will be built using FastAPI, a high-performance Python framework known for its async capabilities and rapid development.
- A secure PostgreSQL database hosted on AWS RDS will manage all structured data, including appointments, patient profiles, and system logs.
- AWS Lambda functions will be used for stateless backend logic, such as sending notifications, validating insurance, and handling slot availability without provisioning persistent servers.

Frontend Interface

- The admin interface for clinic staff will be developed using React.js, designed to offer a clean and responsive user experience.
- UI styling will be handled using TailwindCSS, ensuring fast customizations and mobile responsiveness.
- The project will use Vite as the frontend build tool for optimized loading and fast rendering performance.

DevOps & Cloud Infrastructure

- All services will be deployed on AWS, using API Gateway to manage and secure API access.
- S3 will be used for storing call transcripts, logs, and static assets.
- IAM will enforce role-based permissions, and VPC will ensure private, isolated networking between services.
- CloudWatch will handle monitoring and logging of all backend and voice-related services.

Security Architecture

- To meet healthcare security standards, all data in transit and at rest will be encrypted using TLS 1.3 and AES-256 encryption.

- OAuth 2.0 will be implemented to manage secure authentication and authorization across users and services.
- Session management and audit trails will ensure accountability and traceability for sensitive actions.

Compliance & Risk Monitoring

- The entire infrastructure will follow HIPAA, HITRUST, and SOC2 guidelines.
- Automated security monitoring will be enabled using CloudSploit and ScoutSuite, providing continuous vulnerability scanning, policy enforcement, and compliance auditing.

Weekly Phase Breakdown and Cost

Security & Compliance

- TLS 1.3, AES-256 encryption (at rest & in transit)
- OAuth2.0 for all API access
- Audit logs, access control matrix
- HIPAA, HITRUST, SOC2 ready via AWS services
- Security audits using 3rd-party tools (Cloudsploit, ScoutSuite)

Value Proposition

- Built from scratch to match agentic workflow with Lex/Bland.AI
- Multi-intent, low-latency, multilingual call assistant
- Full compliance with US healthcare data laws
- Scalable SaaS model with clinic self-service
- Best-in-class NLP and speech synthesis with AWS